

Electricidad I:

Funciones de singularidad

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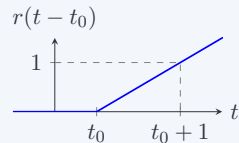
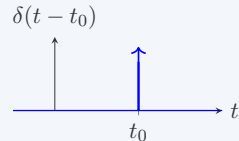
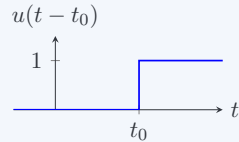
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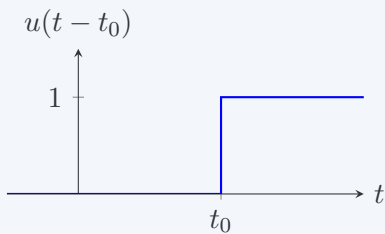
Funciones de singularidad

- La función, su derivada o ambas son discontinuas
- Permiten modelar matemáticamente operaciones de conmutación

Función	Nombre
$u(t - t_0)$	escalón unitario o escalón de Heaviside
$\delta(t - t_0)$	impulso unitario o delta de Dirac
$r(t - t_0)$	rampa unitaria

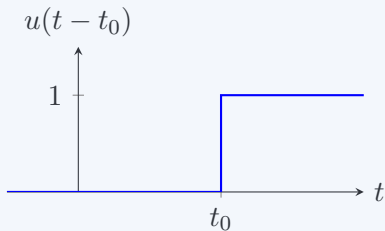


Función escalón unitario



$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

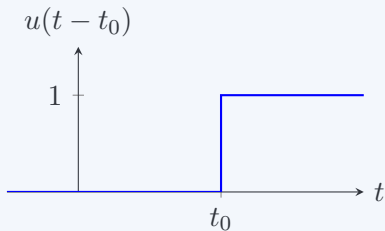
Función escalón unitario



$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

$$\frac{d}{dt}u(t - t_0) = \delta(t - t_0)$$

Función escalón unitario

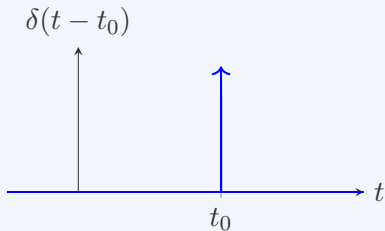


$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

$$\frac{d}{dt}u(t - t_0) = \delta(t - t_0)$$

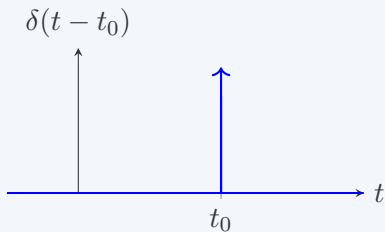
$$\int_{-\infty}^t u(t - t_0) dt = r(t - t_0)$$

Función impulso unitario



$$\delta(t - t_0) = \begin{cases} 0, & t \neq t_0 \\ \infty, & t = t_0 \end{cases}$$

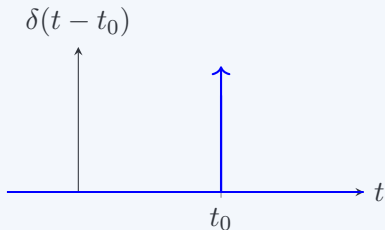
Función impulso unitario



$$\delta(t - t_0) = \begin{cases} 0, & t \neq t_0 \\ \infty, & t = t_0 \end{cases}$$

$$\int_{t_0^-}^{t_0^+} \delta(t - t_0) dt = 1$$

Función impulso unitario

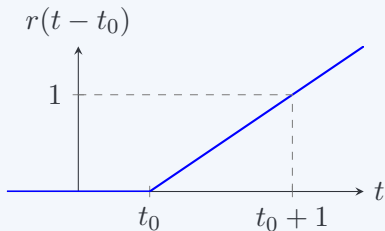


$$\delta(t - t_0) = \begin{cases} 0, & t \neq t_0 \\ \infty, & t = t_0 \end{cases}$$

$$\int_{t_0^-}^{t_0^+} \delta(t - t_0) dt = 1$$

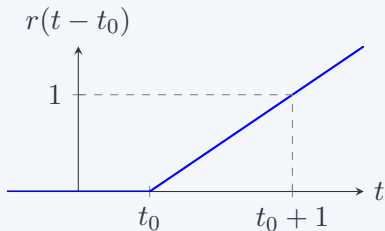
$$\int_{-\infty}^t \delta(t - t_0) dt = u(t - t_0)$$

Función rampa unitaria



$$r(t - t_0) = \begin{cases} 0, & t \leq t_0 \\ t - t_0, & t \geq t_0 \end{cases}$$

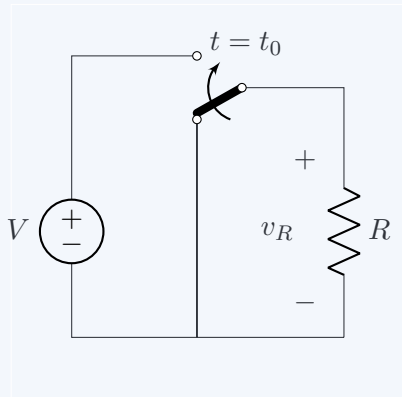
Función rampa unitaria



$$r(t - t_0) = \begin{cases} 0, & t \leq t_0 \\ t - t_0, & t \geq t_0 \end{cases}$$

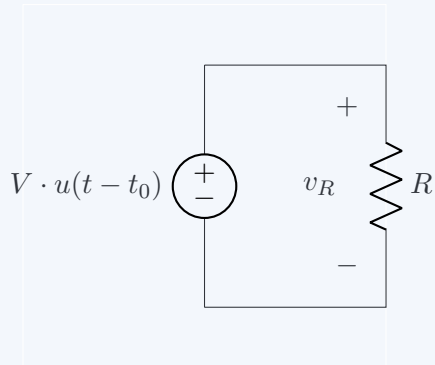
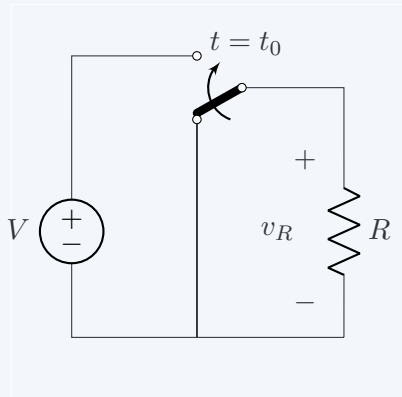
$$\frac{d}{dt}r(t - t_0) = u(t - t_0)$$

Conmutación de fuente de voltaje

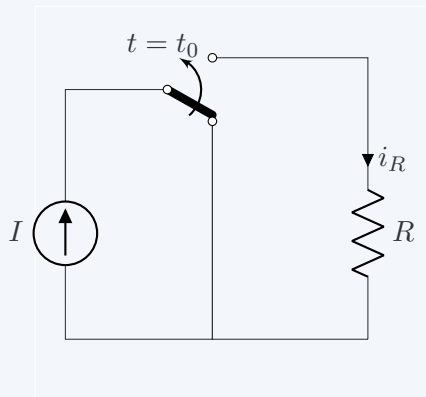


$$v_R = \begin{cases} 0, & t < t_0 \\ V, & t > t_0 \end{cases}$$

Conmutación de fuente de voltaje

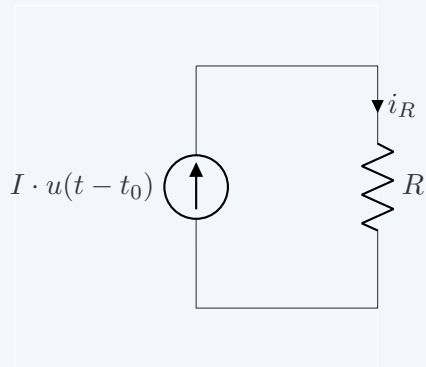
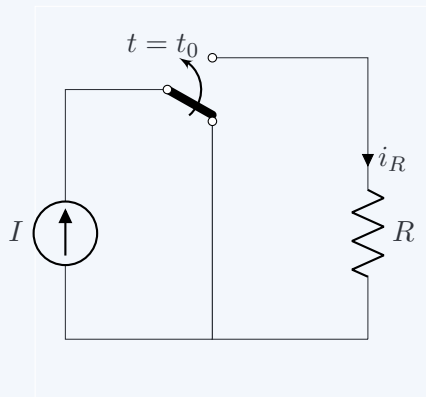


Conmutación de fuente de corriente

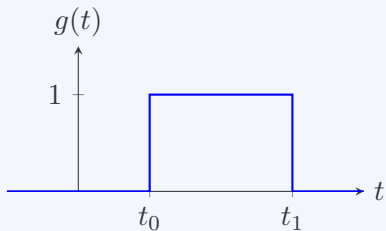


$$i_R = \begin{cases} 0, & t < t_0 \\ I, & t > t_0 \end{cases}$$

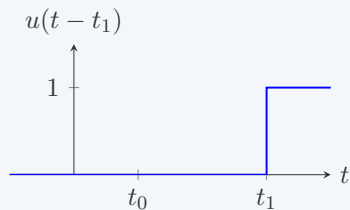
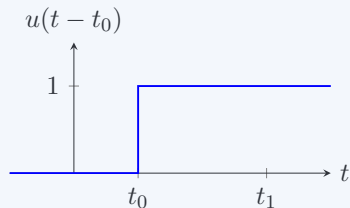
Conmutación de fuente de corriente



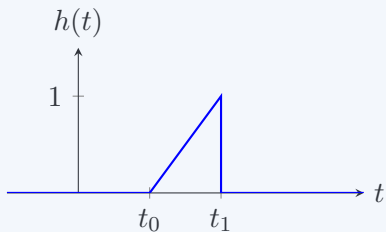
Ejemplos: compuerta



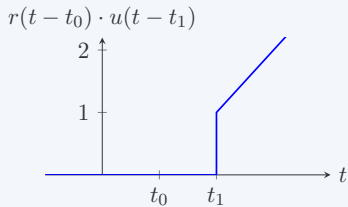
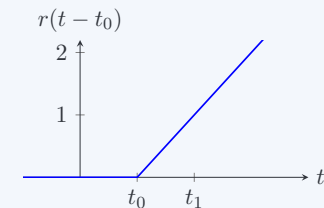
$$g(t) = u(t - t_0) - u(t - t_1)$$



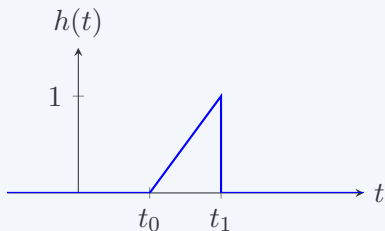
Ejemplos: triangulo



$$\begin{aligned}h(t) &= r(t - t_0) \cdot (1 - u(t - t_1)) \\ &= r(t - t_0) - r(t - t_0) \cdot u(t - t_1)\end{aligned}$$



Ejemplos: triangulo (de nuevo)



$$\begin{aligned} h(t) &= r(t - t_0) \cdot (1 - u(t - t_1)) \\ &= r(t - t_0) \cdot u(-t + t_1) \end{aligned}$$

